Gideon Kofi Amo Appau

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Research interests: continual and long-horizon reinforcement learning; world models and model-based RL; learning under data scarcity (long-tailed recognition, robust optimization).

Education

University of Science and Technology of China (USTC) Suzhou, China
M.S. in Software Engineering (AI/ML concentration); CGPA 3.13/4.3 *Sep. 2024 – Jul. 2027 (Expected)*

- Thesis (working title): *Class-Adaptive Sharpness-Aware Minimization for Long-Tailed Visual Recognition.* Advisor: Assoc. Res. Pengkun Wang.
- Relevant coursework: Artificial Intelligence, Optimization Theory, Information Theory and Coding.

University of Education, Winneba Ghana
B.Sc. in Information Technology Education; CGPA 3.61/4.0 *Aug. 2018 – Oct. 2022*

- First Class Honours, Top 10%. Advisor: Dr. William Asiedu.

Research Experience

University of Science and Technology of China Suzhou, China
Researcher (M.S. thesis); Advisor: Assoc. Res. Pengkun Wang *2025 – Present*

- Proposed **CA-Focal-SAM**, a class-adaptive sharpness-aware minimization optimizer for long-tailed recognition: improves tail-class accuracy by 6.03% on CIFAR-10-LT (IR=200) with no architectural change and zero inference overhead.
- Designed **GammaAdaptor** (2 learnable scalars) mapping class frequency to per-class focal parameters, plus a quadratic curriculum on the SAM perturbation radius; flatter minima verified by Hessian spectral analysis ($\lambda_{\max}$ 114.7 vs. 121.4 for Focal-SAM, stochastic Lanczos quadrature).
- Mechanism study on CIFAR-100-LT (IR 50/100/200; CE, Logit Adjustment, LDAM outer losses) isolated a frozen log-scarcity prior as the component carrying the gain: +2.50 pp tail accuracy over Focal-SAM at IR=100 and +1.98 pp at IR=200 (4 fixed seeds each).
- Built SFCA, an online controller reallocating SAM regularization from live geometric feedback, and tested it with a pre-specified falsification protocol (polarity under two outer losses, signal-destruction controls, hindsight-oracle bound, class-level regression). The feedback carried no measurable information; destroyed feedback matched intact feedback and the oracle lost to the frozen prior.
- Reproducible pipeline for ∼200 training runs: single launcher with a uniform recipe, resume logic, per-run JSON and per-class CSV results, fixed seeds.
- Evaluated on CIFAR-10/100-LT (mean ± std over 3 runs); ImageNet-LT and Places-LT evaluations in progress. Analyses include t-SNE embeddings, 2D/3D loss landscapes, and confusion matrices.

Manuscripts

Appau, G. K. A., & Wang, P. (2026). *Class-Adaptive Sharpness-Aware Minimization for Long-Tailed Visual Recognition* (working title). Manuscript in preparation.

Selected Projects

RL Research Portfolio (github.com/Amo-Gideon/rl-portfolio) *2026 – Present*

- **World model + planning:** on a PyBullet 2-link robot arm, a learned world model (MLP ensemble) with CEM planning reaches 100% single-target success vs. 82% for PPO, and holds 100% under out-of-distribution physics shifts (doubled link mass, 1.5× gravity, halved torque limits) that reduce PPO to 70–80%.
- **Obstacle avoidance:** adding obstacle clearance as a CEM cost cut collision rate from 57% to 23%, where five PPO training configurations failed.
- **Foundations:** reproduced Dueling/Double DQN from scratch; built an end-to-end RLHF pipeline (SFT, reward model, PPO alignment) for a 0.5B-parameter language model.

LLM Application Projects *2025*

- **Personalized Agentic RAG** (USTC): adaptive retrieval with autonomous agents, iterative self-correction, and user memory (LangGraph, LangChain, FAISS).
- **Campus Knowledge Base RAG Chatbot:** semantic campus-information retrieval with a Chroma vector database (Ollama, Gradio).
- **AI Study Coach:** multi-agent assistant with 4 AutoGen agents and Gemini 2.0 Flash for adaptive quiz generation.

Industry Experience

Future Interns Remote
Machine Learning Intern *Mar. 2026 – Apr. 2026*

- Built a 6-month sales forecasting system and a BERT/DistilBERT ticket-classification pipeline with sentiment analysis for priority routing.
- Developed an AI recruitment tool combining spaCy/NLTK resume parsing with LLM-based job matching.

Akenten Appiah Menkah University Kumasi, Ghana
IT Assistant *Nov. 2022 – Aug. 2023*

- Automated data workflows in Python (Pandas, NumPy), cutting manual processing time by 40%; optimized SQL databases (5,000+ records), improving query performance by 30%.

Amidarr A-mole Ghana
Full-Stack Web Developer Intern *Aug. 2021 – Mar. 2022*

- Built 3 full-stack web applications (Flask, JavaScript, MySQL), integrating the PayStack payment API into an e-commerce platform.

Teaching, Mentorship, and Presentations

Peer Mentor, USTC Software Engineering: guided junior master's students on PyTorch and SAM *2024 – 2025*
Talk: USTC Thesis Progress Seminar, *CA-Focal-SAM for Long-Tailed Visual Recognition* *2026*
Talk: USTC Engineering Practice Defense, *Homomorphic Encryption Deployment on FPGA* *2025*

Honors and Certifications

Chinese Government Scholarship (CSC), USTC *2024*
NVIDIA, LLM RAG Agent Fundamentals *2025*
DeepLearning.AI, Advanced Learning Algorithms *2025*
DeepLearning.AI, Supervised Machine Learning: Regression & Classification *2025*

Skills

Languages:	Python, JavaScript, SQL, Java, Bash
RL:	PPO, Dueling/Double DQN, RLHF, CEM planning, world models (PyBullet)
ML/Optimization:	PyTorch, scikit-learn, sharpness-aware minimization, Hessian spectral analysis
LLM/NLP:	Transformers, BERT, DistilBERT, spaCy, NLTK, LangChain, LangGraph, AutoGen, Ollama, FAISS, Chroma
Data & Tools:	Pandas, NumPy, Matplotlib, Flask, REST APIs, MySQL, Git, Linux, Docker, LaTeX

References: available upon request.